

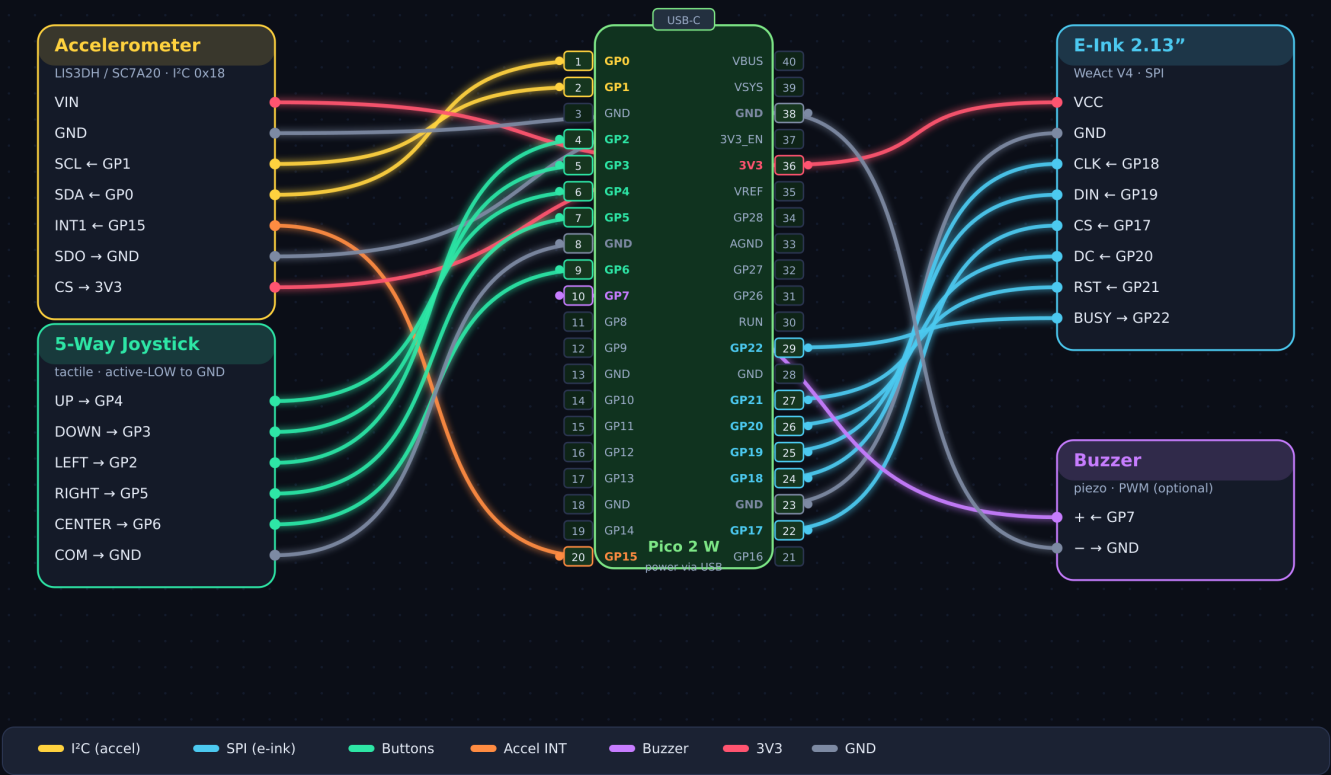
# Breadboard Wiring Guide

Prototype the Dilder on a breadboard to test firmware before committing to the PCB. Pin-for-pin identical to the board's GPIO map — flash the same binary, no code changes.

MCU Raspberry Pi Pico 2 W

Bus I<sup>2</sup>C0 @ 0x18 · SPI

Power USB 5V → onboard 3V3



## BILL OF MATERIALS

| Qty | Part                                         | Notes                                                            |
|-----|----------------------------------------------|------------------------------------------------------------------|
| 1   | Raspberry Pi Pico 2 W (with headers)         | same MCU as the board · power & flash over USB                   |
| 1   | LIS3DH / SC7A20-compatible accel breakout    | register-compatible · set address pin LOW → <b>0x18</b>          |
| 1   | WeAct 2.13" e-paper module (V4)              | 8-pin SPI · BUSY/RST/DC/CS/CLK/DIN/GND/VCC                       |
| 5   | tactile push-buttons (or a 5-way nav module) | UP / DOWN / LEFT / RIGHT / CENTER                                |
| 2   | 2.2 kΩ resistors                             | I <sup>2</sup> C pull-ups (skip if your accel breakout has them) |
| 1   | passive piezo buzzer (optional)              | GP7 PWM · sound test only                                        |
| 1   | breadboard + jumper wires                    | 830-point + M-M / M-F jumpers                                    |

## EXACT CONNECTIONS

| ACCELEROMETER — I <sup>2</sup> C0 |        |            |
|-----------------------------------|--------|------------|
| Pico pin                          | wire   | module pin |
| GP0 (pin 1)                       | ● SDA  | SDA        |
| GP1 (pin 2)                       | ● SCL  | SCL        |
| GP15 (pin 20)                     | ● INT  | INT1       |
| 3V3 (pin 36)                      | ● pwr  | VIN/VCC    |
| GND (pin 38)                      | ● gnd  | GND        |
| GND                               | ● addr | SD0/SA0    |
| 3V3                               | ● mode | CS         |
| JOYSTICK — active-LOW             |        |            |
| Pico pin                          | wire   | module pin |
| GP4 (pin 6)                       | ● sig  | UP         |
| GP3 (pin 5)                       | ● sig  | DOWN       |
| GP2 (pin 4)                       | ● sig  | LEFT       |
| GP5 (pin 7)                       | ● sig  | RIGHT      |
| GP6 (pin 9)                       | ● sig  | CENTER     |
| GND (pin 8)                       | ● gnd  | COMMON     |

| E-INK 2.13" — SPI |        |            |
|-------------------|--------|------------|
| Pico pin          | wire   | module pin |
| GP17 (pin 22)     | ● CS   | CS         |
| GP18 (pin 24)     | ● CLK  | CLK/SCL    |
| GP19 (pin 25)     | ● MOSI | DIN/SDA    |
| GP20 (pin 26)     | ● DC   | DC         |
| GP21 (pin 27)     | ● RST  | RST/RES    |
| GP22 (pin 29)     | ● BUSY | BUSY       |
| 3V3 (pin 36)      | ● pwr  | VCC        |
| GND (pin 23)      | ● gnd  | GND        |
| BUZZER — optional |        |            |
| Pico pin          | wire   | module pin |
| GP7 (pin 10)      | ● PWM  | +          |
| GND (pin 38)      | ● gnd  | –          |

### GOTCHAS

- I<sup>2</sup>C address:** tie the accel's SDO / SA0 pin to **GND** so it answers at **0x18** (the firmware's expected address). CS → 3V3 keeps it in I<sup>2</sup>C mode.
- Pull-ups:** one pair of 2.2 kΩ from SDA and SCL to 3V3 — only one pair on the whole bus. Most accel breakouts already have them; don't double up.
- Buttons are active-LOW:** each switch shorts its GPIO to GND when pressed. Enable the Pico's internal pull-ups in firmware — no external resistors needed.
- E-ink SPI:** CLK=GP18, DIN(MOSI)=GP19. The panel is write-only except BUSY (an input back to GP22). 3.3 V logic throughout — no level shifting.
- Power:** just plug USB into the Pico — its onboard regulator feeds the 3V3 rail. Skip the TP4056 / battery / slide-switch entirely for firmware bring-up.

### BRING-UP CHECKLIST

- ✓ Hold BOOTSEL, plug in USB, drag-drop the firmware — board enumerates as RPI-RP2.
- ✓ I<sup>2</sup>C scan finds the accelerometer at 0x18 on GP0/GP1.
- ✓ Accel returns live X/Y/Z; tilting changes values; INT1 (GP15) fires on motion.
- ✓ E-ink refreshes a test pattern (confirms GP17-22 SPI).
- ✓ Each of the 5 directions reads LOW when pressed (GP2-6).
- ✓ Buzzer tone plays on GP7 (if fitted).